

Cybersecurity Threat Analytics Using CICIDS2017 Dataset

Network Traffic Analysis and Web Attack Detection Using Python, SQL and Power BI

PYTHON

PANDAS

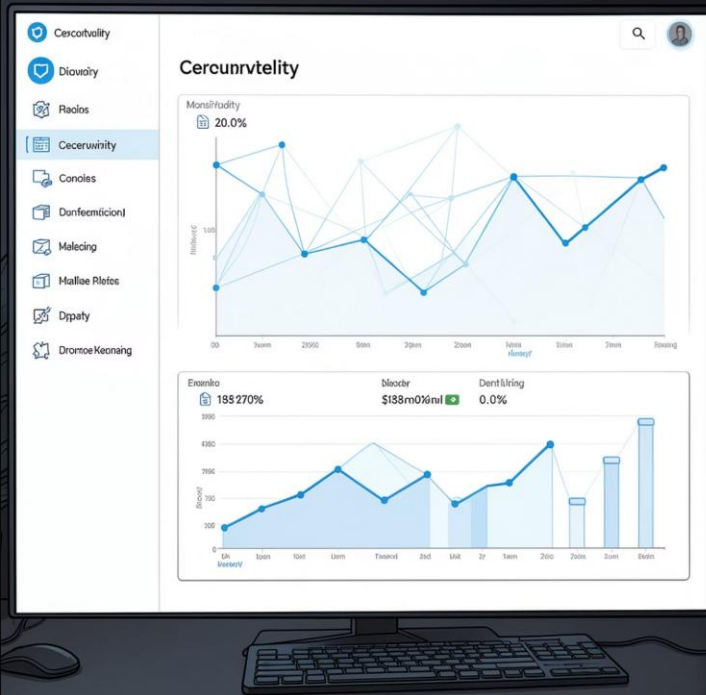
NUMPY

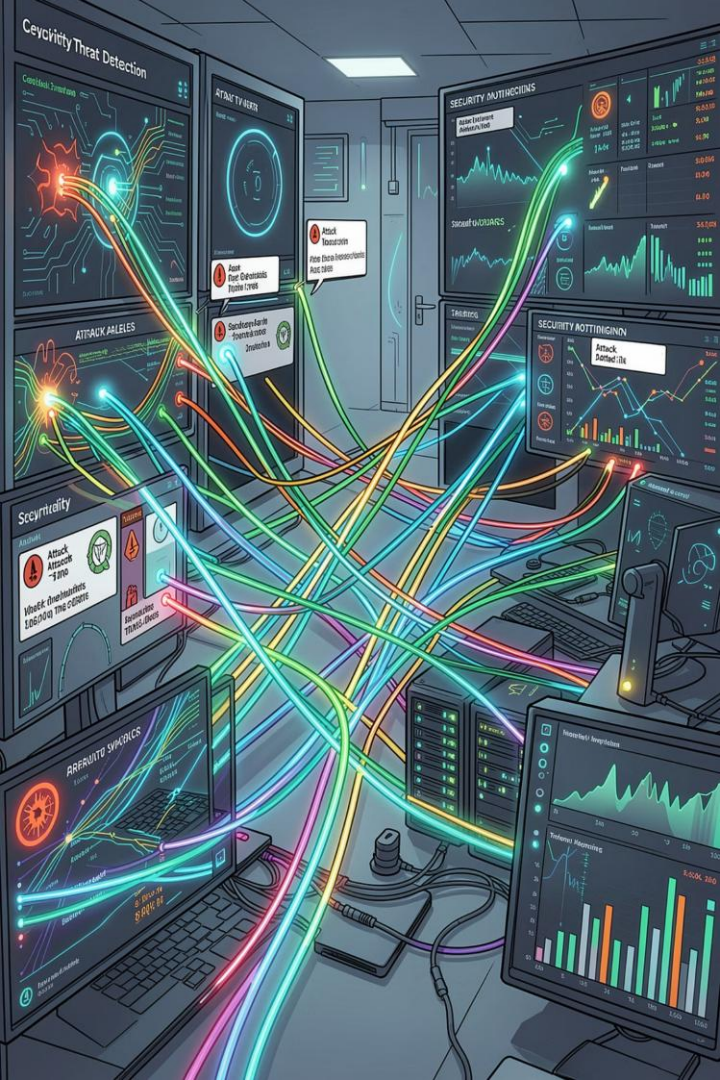
MYSQL

MATPLOTLIB

SEABORN

POWER BI





The Challenge: Detecting Threats at Scale

Modern organizations generate **millions of network events daily**. Identifying malicious traffic within large volumes of network logs is increasingly complex — requiring systematic analysis, intelligent tooling, and actionable dashboards.

Detect Attack Patterns

Identify signatures of known web attacks across traffic flows

Analyze Traffic Behavior

Profile normal vs. anomalous network communication patterns

Identify Suspicious IPs

Flag high-risk source and destination entities

Build Security Dashboards

Generate visual insights for rapid decision-making

Dataset Overview: CICIDS2017

This project uses the **CICIDS2017 – Thursday Working Hours Web Attacks** dataset, a widely recognized benchmark for network intrusion detection research.

170,000+ Records

Network flow events captured during realistic web attack scenarios

4 Attack Classes

BENIGN, Brute Force, XSS, and SQL Injection

8 Feature Categories

Source/Dest IP, Ports, Protocol, Flow Duration, Packet Stats, TCP Flags, Attack Labels

BENIGN



BRUTE FORCE



XSS



SQL INJECTION



Data Cleaning & Preprocessing

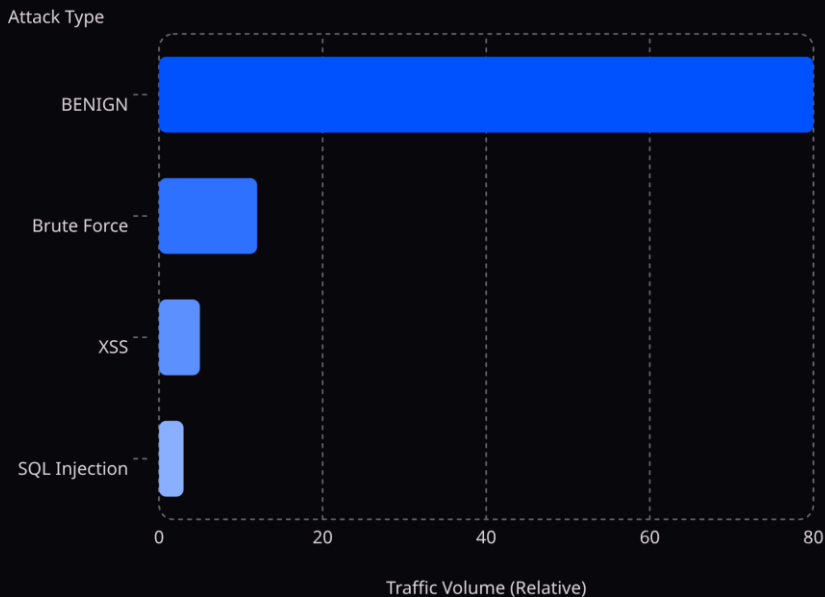
A rigorous preprocessing pipeline ensured data quality and consistency before analysis — removing noise, fixing schema issues, and standardising features for reliable threat modelling.



✓ Outcome: A clean, structured dataset ready for exploratory analysis, SQL querying, and dashboard visualisation.

Exploratory Data Analysis

EDA revealed critical patterns in attack distribution, protocol behaviour, and traffic characteristics — forming the analytical foundation for threat detection.

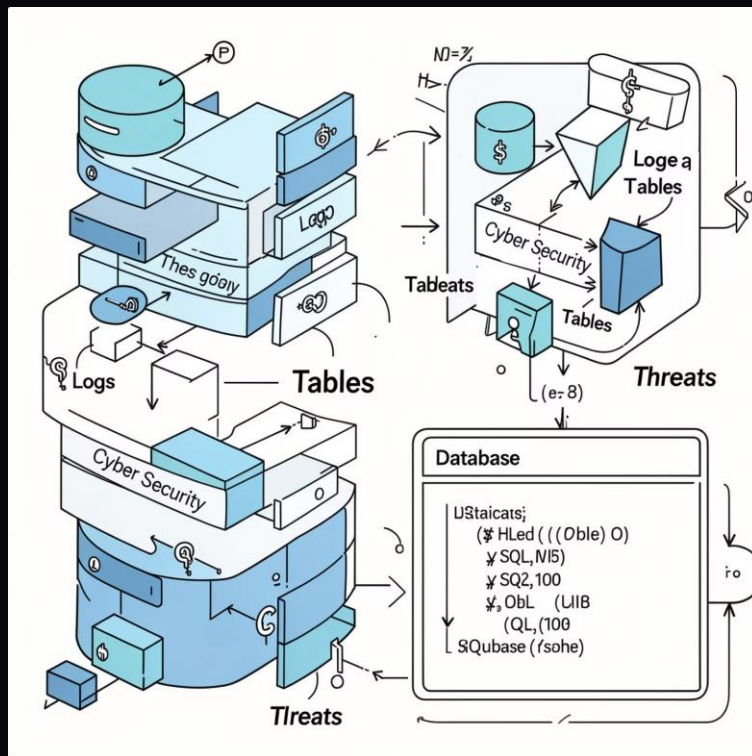


Key EDA Findings

- **Brute Force** attacks represented the largest share of malicious traffic
- Protocol usage showed near-equal TCP/UDP distribution
- XSS attacks exhibited **longer average flow durations**
- Distinct source IP clustering patterns emerged for each attack type

SQL-Based Threat Analysis

Structured SQL queries enabled deep-dive analysis across attack distribution, IP behaviour, port utilisation, and risk scoring — turning raw logs into threat intelligence.



→ Attack Distribution

Grouped counts by attack label to quantify threat landscape

→ Source & Destination IP Analysis

Identified top talkers and high-frequency suspicious entities

→ Port & Protocol Profiling

Mapped attack activity to specific service ports

→ Risk Scoring

Scored IPs using TCP flag patterns and flow characteristics



Key Finding: A small number of IP addresses generated a disproportionately large share of malicious network activity.

CYBER SECURITY

LOGS

TABLES

THREATS

Key Security Insights

1

Brute Force Dominance

Accounted for the majority of all detected malicious activity

2

Port 53 & 443 Targeted

DNS and HTTPS ports were heavily utilised across attack types

3

TCP $\approx$ UDP Distribution

Traffic was almost equally split between TCP and UDP protocols

4

XSS Flow Duration

XSS attacks showed significantly longer average flow durations

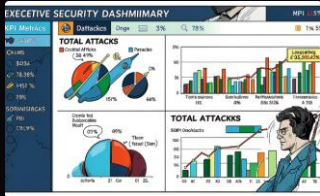
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High-Risk IPs Flagged

TCP flag-based scoring identified the most dangerous source IPs

Power BI Threat Analytics Dashboard

A multi-page interactive dashboard was developed to translate analytical findings into operational security intelligence for SOC teams and stakeholders.



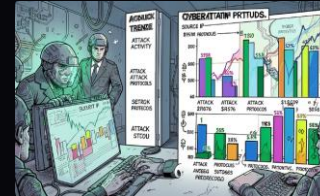
Executive Overview

Total records, attack counts, and attack percentage at a glance



Traffic Analysis

Protocol distribution, flow duration trends, and packet statistics



Attack Analysis

Attack trends, source IP activity heatmaps, and attack protocol breakdowns

Security Recommendations

1

Enable MFA

2

Rate limiting

3

Monitor high-risk
IPs

4

Deploy IDS/IPS

5

Secure web apps

6

Monitor DNS & HTTPS

✓ **Expected Benefits:** Reduced attack surface, faster threat detection, and improved overall security posture.

Conclusion & Future Scope

This project successfully analysed network traffic from the CICIDS2017 dataset, identifying malicious patterns and producing actionable security intelligence using Python, SQL, and Power BI.

Achievements

- Cleaned and preprocessed 170K+ network flow records
- Performed SQL-based threat analytics and risk scoring
- Generated visual insights via Matplotlib and Seaborn
- Built an interactive multi-page Power BI dashboard
- Produced prioritised, actionable security recommendations

Future Scope

01

Real-Time Threat Monitoring

Stream processing for live network traffic analysis

02

Machine Learning Detection

Supervised and unsupervised models for anomaly detection

03

SIEM Integration

Connect findings to enterprise security information platforms

04

Cloud Security Analytics

Extend methodology to cloud-native traffic and log sources